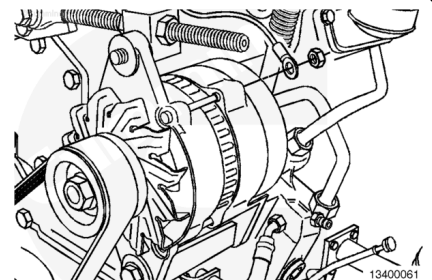


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



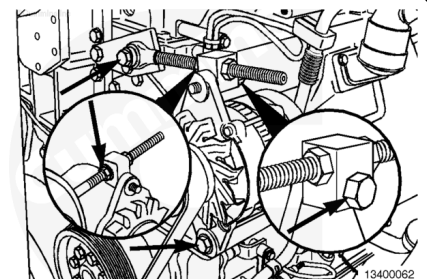
Disconnect the batteries.

Disconnect the wiring and ground strap from the alternator.

Loosen the adjusting link and the alternator mounting capscrews.

Loosen the adjusting nut to remove tension on the alternator belt.

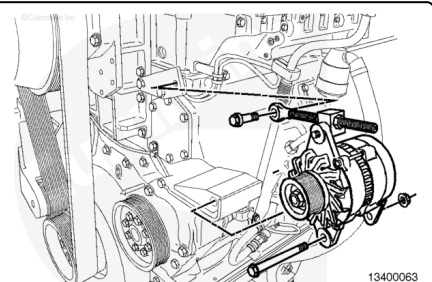
Remove the alternator belt.



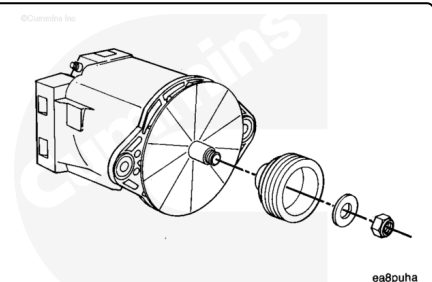
Remove the capscrew that fastens the adjusting link to the alternator bracket.

Remove the lower alternator mounting capscrew and nut.

Remove the alternator and adjusting link.



Remove the nut and the pulley from the alternator.



Test

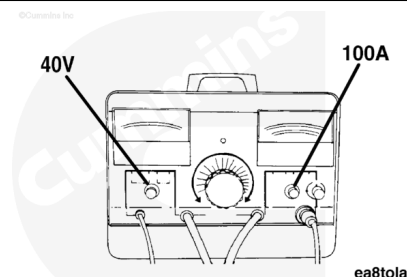
The following instructions are for use with the inductive charging and cranking system analyzer or equivalent.

Before performing the following test, be sure the alternator belt is tightened to the correct specifications. Refer to Procedure 013-005 in Section 13.

(/qs3/pubsys2/xml/en/procedures/89/89-013-005.html).

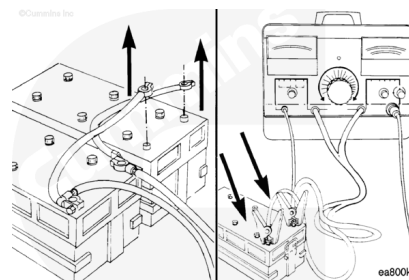
Set the voltage selector knob to the appropriate scale. For a 24-VDC system, choose the 40-VDC scale.

Set the amp selector knob to 100 amps.



⚠ WARNING ⚠

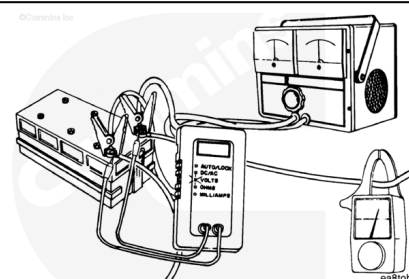
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Remove the cables to any other battery in the circuit.

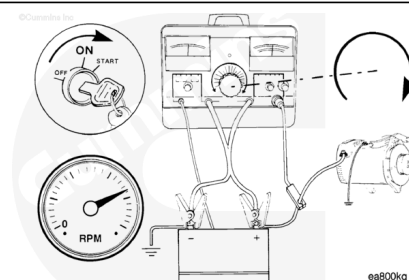
Connect the correct analyzer leads to the positive and negative terminals on the battery.

Connect the clamp-on amp pick-up to the alternator output cable as far away from the alternator as possible.



Operate the engine at high idle and turn the analyzer load control knob **clockwise** until a maximum amp reading is obtained.

Do **not** let the load voltage drop below 26- VDC for a 24-VDC system.

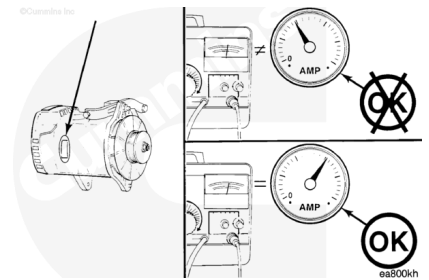


The maximum amp reading is the alternator output and **must** meet the alternator manufacturer's specifications.

The alternator maximum rated output is normally stamped or labeled on the alternator.

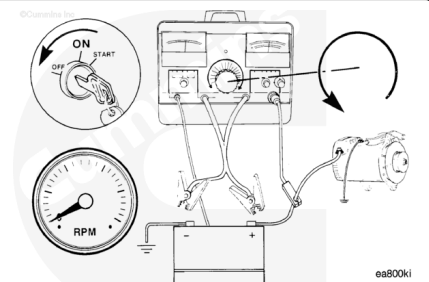
Also check the equipment ammeter gauge. If it does **not** read approximately the same as the test equipment, it should be replaced.

If the alternator output is **not** within 10 percent of rated output, repair or replace the alternator. Refer to the manufacturer's instructions.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Turn the analyzer load control knob **counterclockwise** to the OFF position and shut off the engine.

Remove the test equipment. Connect all battery cables that were removed.

Install

Engines with 75A and 140A Alternator:

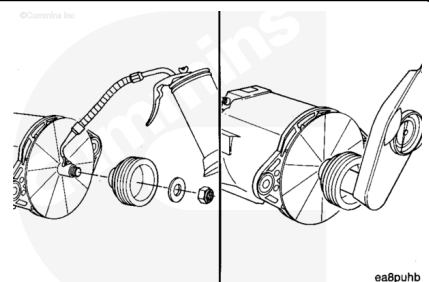
Clean and check the pulley for reuse. Refer to Procedure 013-006 (</qs3/pubsys2/xml/en/procedures/89/89-013-006.html>).

Lubricate the shaft with clean 15W-40 engine oil.

Install the pulley and nut on the alternator shaft.

Tighten the nut.

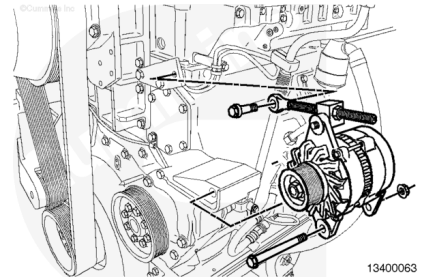
Torque Value: 100 n•m [75 ft-lb]



Engines with 75A Alternator:

⚠ CAUTION ⚠

Do not attempt to pry the belt on the pulley to avoid damage to the pulley and belt.



Install the alternator and the adjusting link as shown.

Leave the alternator mounting capscrews loose until the belt is adjusted.

Install the belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/89/89-013-005.html)

Tighten the jam nut on the adjusting link.

Tighten the alternator mounting capscrews.

Torque Value:

Jam nut

1. 196 n•m [125 ft-lb]

Torque Value:

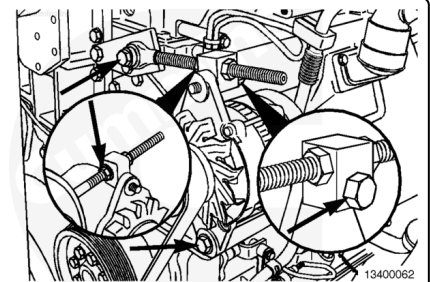
Alternator mounting capscrew (Top)

1. 75 n•m [55 ft-lb]

Torque Value:

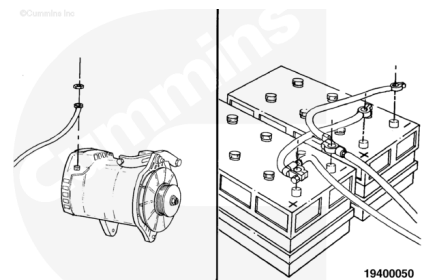
Alternator mounting capscrew (Bottom)

1. 120 n•m [89 ft-lb]



⚠ WARNING ⚠

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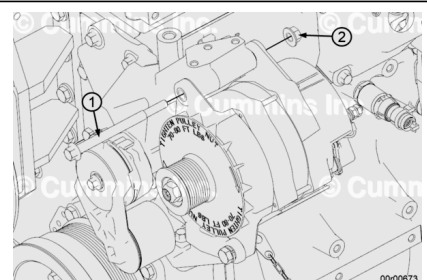
Connect the wiring to the alternator.

Connect the batteries.

Engines with 140A Alternator:

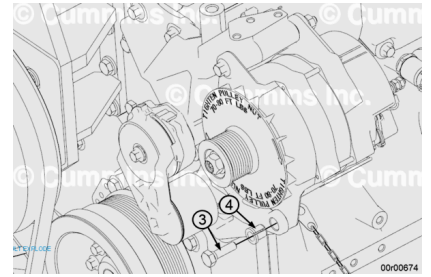
Install the alternator on the alternator mounting bracket with capscrew (1) and nut (2).

Torque Value: 46 n•m [34 ft-lb]



Engage capscrew (3) and bushing (4).

Torque Value: 71 n•m [52 ft-lb]



Last Modified: 24-Aug-2020
